

SWAYAM MEHTA

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Applied ML Engineer | LLM Systems | RAG | ML Infrastructure

EDUCATION

MS Computer Science	2024 - 2026
Arizona State University, Ira A. Fulton Schools of Engineering	4.0 GPA
BTech Computer Science and Engineering	2020 - 2024
Vellore Institute of Technology	8.65 CGPA

SKILLS

ML/AI	PyTorch, Transformers, LangChain, LangGraph, RAG, vLLM, spaCy, NER, CUDA
Languages	Python, SQL, Java, JavaScript, Bash
Data/Infra	Pandas, NumPy, PostgreSQL, Neo4j, ChromaDB, AWS, Kubernetes, Prometheus, Grafana, Falco
Web/Tools	FastAPI, Flask, React, Next.js, Git, Linux, Docker

EXPERIENCE

JobTalk AI: Machine Learning Engineer Intern	03/2026 - 05/2026
- Designed a resume extraction pipeline combining regex, spaCy NER with an ESCO occupational-skills ontology, and LLM structured outputs merged via a per-field trust hierarchy, reaching 0.72 F1 across 8 fields on a hand-labeled evaluation set. - Cut LLM inference cost 73% and eliminated extraction timeouts (50% → 0%) by consolidating 5 per-section prompts into one schema call, self-hosted on a quantized Qwen3-8B via vLLM. - Kept all candidate PII off third-party APIs by self-hosting the extraction model instead of routing through external LLM providers.	
Vellore Institute of Technology: Research Assistant	01/2024 - 05/2024
Improved GAN-based stock price prediction models by incorporating market indices and technical indicators as additional features, reducing prediction error (RMSE) by 40% over prior research baselines.	
Hewlett Packard Enterprise: CTY Intern - High Performance Computing	01/2023 - 06/2023
- Engineered Kubernetes runtime security monitoring using Falco (eBPF), Prometheus and Grafana, enabling real-time detection of container escapes, privilege escalation and filesystem tampering. - Boosted runtime security coverage (~30%) and slashed incident detection latency by (~40%) in controlled attack simulations by replacing manual log inspection with rule-based behavioral alerts.	

PROJECTS

ClickLess AI [Python, LangGraph, Mistral 7B, Neo4j, GraphRAG, Ollama, Apriori]	Spring 2026
- Built a grocery shopping agent that converts natural-language requests into ranked product options with dietary-constraint filtering, using Mistral 7B to extract SPO triples from Open Food Facts into a Neo4j graph of 8,146 products and 43,216 edges. - Implemented a weighted ensemble ranker combining Apriori co-purchase rules (116 rules over 100K baskets) with KG dietary-constraint scoring, reaching 75% Triple Precision and improving NDCG@5 from 0.58 to 0.77 over a classical-only baseline.	
LLM-Assisted FMEA in SMT [Python, LangChain, HuggingFace, ChromaDB, BM25]	Spring 2025
- Built a RAG pipeline over semiconductor surface-mount manufacturing data to auto-generate Failure Mode and Effects Analysis (FMEA) quality-control tables, achieving 0.78–0.93 semantic similarity, 90% field coverage, and 80% reduction in manual analysis effort. - Elevated BLEU/ROUGE scores by 15–20% over baseline LLMs via hybrid retrieval (BM25 + bge-small-en embeddings), enabling production-ready FMEA generation with full context attribution and minimal expert edits.	

LEADERSHIP

IEEE Computer Society - VIT: Secretary	01/2023 - 12/2023
Led a cross-functional team to launch Chota Dhobi, a laundry-tracking app with 24K+ active users, driving 15% month-over-month growth through targeted onboarding campaigns.	